

Natural Language Processing

Project 2.1: From Classical ML to LLMs in NLP

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Abstract

Multilingual ADE detection in medical social media is studied (English/Russian training; German zero-shot), and classical ML, Transformers, and LLM prompting are compared on structured Iris data. Cross-lingual transfer, class imbalance, and practical trade-offs across modeling paradigms are analyzed.

Keywords

Multilingual NLP, ADE detection, Transformers, Cross-lingual transfer, LLM, Iris

1 Introduction

Two Tasks are addressed: (i) multilingual ADE detection in noisy social media under severe class imbalance and domain shift, and (ii) classification on small tabular data, where classical ML, Transformers on textified inputs, and LLM prompting are compared.

2 Task 1: Multilingual Transfer and Translation in Medical Social Media NLP

This task involves the binary classification of user-generated social media posts across multiple languages to determine whether a post contains a personal mention of an Adverse Drug Event (ADE). The dataset presents significant natural language processing challenges, including severe class imbalance, the noisy and informal nature of social media text, and considerable variations in sequence length across different languages. Our approach focuses on leveraging pre-trained transformer models to address these issues and evaluate cross-lingual generalization.

2.1 Task 1.1: Language Selection, Data Preparation, and Splits

For this task, three languages were selected to investigate cross-lingual transfer capabilities in medical social media analysis:

- **English (en):** Selected as the primary high-resource source language.
- **Russian (ru):** Selected as a secondary source language. Including Russian adds linguistic diversity (e.g., Cyrillic script, rich morphology), preventing the model from overfitting to Germanic/Latin linguistic patterns.
- **German (de):** Selected as the **Zero-Shot Target** language. It represents a low-resource scenario in this dataset with a very low number of positive cases. This makes it an ideal candidate for evaluating how well models trained on English and Russian generalize to unseen languages.

The training dataset was partitioned into an 80% training set and a 20% validation set. To ensure strict reproducibility and consistency across all experiments, the resulting splits were saved as static CSV

files (train.csv and val.csv). This guarantees that hyperparameter tuning, model training, and final evaluations are conducted on the exact same data distributions, effectively eliminating the risk of data leakage between experimental runs.

2.1.1 Dataset Statistics & Class Imbalance: A major challenge identified in the dataset is severe **class imbalance**. As shown in Table 1, the ratio of positive samples (Adverse Drug Events) is consistently low in all languages.

Language	Total	Pos	Neg	Pos (%)
English (en)	17,128	1,198	15,930	7.0
Russian (ru)	10,695	1,082	9,613	10.1
German (de)	1,482	83	1,399	5.6
TOTAL	29,305	2,363	26,942	8.1

Table 1: Data volume and positive-class rate across subsets.

Observation: Without intervention, the model will likely bias heavily towards the negative class. This issue was addressed during the train/validation split by applying **Stratified Splitting** (stratified by Language + Label). This ensured that the validation set maintained the exact same positive ratios (e.g., English: 7.0%, Russian: 10.1%) as the training set, preventing skewed evaluation results.

2.1.2 Preprocessing: A custom preprocessing pipeline, tailored for Social Media Mining for Health (SMM4H), was implemented. The impact of each step on the training/validation and test datasets is quantified below:

- **Data Cleaning (Deduplication & Empty Rows):** All datasets were evaluated for empty text entries, resulting in 0 removals. Deduplication was subsequently applied to prevent data redundancy. This removed **61** duplicate rows from the train/validation set (initial: 29,305 → final: 29,244) and **3** from the test set (initial: 4,191 → final: 4,188).
- **User Anonymization (@User → [USER]):** Affected **7,942** records in the train/validation set and **570** in the test set. This was the most frequent noise pattern. Anonymization is crucial to strictly adhere to privacy standards and to prevent the model from overfitting to specific active users (i.e., learning that a specific user is sick, rather than learning the symptoms).
- **URL Anonymization:** Hyperlinks were replaced with a generic token to remove irrelevant routing information, affecting **802** records in the train/validation set and **67** in the test set.
- **Hashtag Cleaning (#flu → flu):** Affected **3,287** records in the train/validation set and **180** in the test set. Hashtags often contain the most important semantic content and keywords. Removing the hashtag symbol allows the tokenizer

to treat them as standard words, preserving the medical context.

- **HTML & Artifact Removal:** Affected 2,281 records in the train/validation set and 1,024 records in the test set. Tags like `<br>` disrupt sentence structure. They were replaced with periods to restore clean sentence boundaries for the Transformer model.
- **Emoji Handling (→ :nauseated_face:):** Affected 478 records in the train/validation set and 60 in the test set. Emojis are dense information carriers in social media (e.g., a pill or vomit face). Demojizing them into text ensures the model interprets the semantic sentiment rather than treating them as unknown tokens.
- **Capitalization (Case Preservation):** Original capitalization was explicitly **preserved**. Medical acronyms (e.g., AIDS, ALL, US) are highly case-sensitive. Lowercasing them would introduce semantic ambiguity (e.g., hearing "aids" vs. the disease).

2.1.3 Final Splits: Following preprocessing, the data (comprising en, ru, and de) was partitioned into **Train:** 23,395 samples, **Val:** 5,849 samples, and **Test:** 4,188 samples.

2.1.4 Machine Translation. Automated translation was initially attempted using cloud-hosted models (e.g., Mistral Large) via the Academic Cloud API. However, strict rate limits rendered this approach unscalable for large datasets, capping throughput at approximately 190 samples despite programmed backoff strategies. Consequently, the pipeline was transitioned to local inference using the `mistralai-devstral-small-2-24b-instruct-2512` model, which provided an optimal balance of execution speed and strict instruction adherence.

A total of 1,000 English samples were translated into German across three sequential processing sessions. To ensure methodological consistency for cross-lingual evaluation, the positive class (ADE) ratio of the translated subset was artificially constrained to exactly 8%, mirroring the inherent class imbalance of the primary training dataset.

Translations were generated using a highly constrained zero-shot prompting strategy. The model was assigned a professional medical translator persona with strict rules enforced:

- **Semantic Fidelity:** Exact preservation of the original medical meaning.
- **Acronym Retention:** Preservation of standard medical acronyms (e.g., AIDS, PFS) in their original form.
- **Zero-Tolerance for Artifacts:** Strict prohibition of prefaces (e.g., "Translation:"), conversational fillers, appended notes, extra lines, or quotation marks.

2.2 Task 1.2: Multilingual Data Exploration and Analysis

2.2.1 Dataset Distribution & Balance. An exploratory data analysis was conducted to evaluate the label and language distributions across the defined data splits.

As illustrated in Figure 1, a severe class imbalance is evident across all subsets and languages, with the negative class substantially outnumbering the positive class.

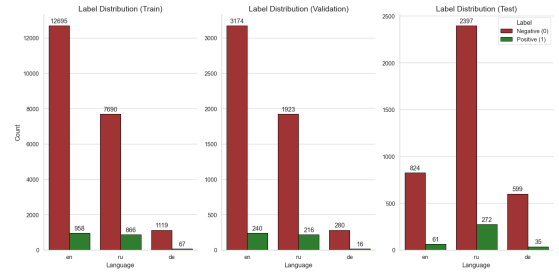


Figure 1: Label distribution across splits.

An analysis of the training corpus reveals a distinct language hierarchy. English serves as the dominant language, comprising **58.4%** (13,653 samples) of the training data, followed by Russian at **36.6%** (8,556 samples). German represents a low-resource minority, accounting for only **5.1%** (1,186 samples). This imbalance indicates that without intervention (e.g., balanced batching or resampling), a multilingual model will inherently bias its internal representations toward English syntax and vocabulary.

Furthermore, a significant distribution shift is observed in the evaluation set. While English dominates the training set, the test set is predominantly Russian (**63.7%**, 2,669 samples), with English dropping to **21.1%** (885 samples) and German increasing to **15.1%** (634 samples).

The origin of the data spans diverse platforms, which directly impacts the linguistic style of the corpus. As detailed in Table 2, the vast majority of the training data consists of short, informal text (tweets), which contrasts with the longer narrative structures found in forum posts.

Source Type	Percent
Tweet	80.8%
Sentence	12.9%
Forum Post	5.1%
Review	1.3%

Table 2: Distribution of text source types within the training dataset.

2.2.2 Linguistic Characteristics & Sequence Lengths. To understand input complexity and structural differences across the dataset, whitespace-split token lengths were analyzed.

- **Cross-Lingual Length Variations:** English and Russian samples exhibit similar, concise average token lengths (**16.58** and **19.24** tokens, respectively). In stark contrast, the German subset is significantly longer, averaging **100.14** tokens (with a maximum of 984 tokens).
- **Discourse Structure Shift:** The drastic length of the German texts strongly suggests they originate from *Forum Posts*

rather than *Tweets*. Consequently, zero-shot cross-lingual transfer to German requires the model to generalize not only across distinct vocabularies but also across a severe shift in discourse structure (i.e., transitioning from short, noisy microblogs to long, narrative paragraphs).

- **Length vs. Label Correlation:** A structural correlation between sequence length and class label was identified. Positive samples (indicating an Adverse Drug Event) are, on average, longer (**31.93** tokens) than negative samples (**20.90** tokens). This suggests that users describing an actual side effect tend to provide more narrative detail and context compared to users making simple, declarative statements about a drug.

2.2.3 Qualitative Analysis and Manual Inspection. A manual inspection of the LLM-generated translations (English to German) was conducted to identify translation artifacts, stylistic shifts, and language-specific phenomena in medical social media text.

Translating: 100% Translation Preview (First 5 samples) 500/500 (2:00:01:00:00, 14.4kb/s)	
Original (EN)	Translated (DE)
how to use paxil for premature ejaculation (LIFENESS & Approved ... how to use paxil for premature ejaculation :: online retailer of ...	Wie man Paxil bei vorzeitigem Samenerguss einsetzt (LIFENESS & ... Online-Händler von ...
Spending the next few hours seeing a Kenneth Paltraw movie. I'll be in the front row doing Prolia arms.	Die nächsten Stunden verbringe ich damit, einen Film von Kenneth Paltraw zu sehen. Ich werde in der ersten Reihe mit dem Prolia-Armen.
[USER] [USER] me too! but it's different pain ya know? my back has hurt for years...it's not like that. i'm already in neurontin	[USER] [USER] das same, eben es ist ein anderer Schmerz, oder? Mein Rücken tut seit Jahren weh... das ist nicht das same. Ich bin schon auf Neurontin.
[USER] Watch your eyesight with Lamictal - I went from awesome vision, to wearing glasses full time for the 2 yrs I was on it.	[USER] Achte auf deine Sehfähigkeit bei Lamictal - Ich ging von einer ausgezeichneten Sicht, zu voller Brillenabhängigkeit für die 2 Jahre, die ich es einnahm.
Phys Rehab Doc: "So why do you take Lamictal?" Me: "Bipolar disorder." Doc: "You don't seem bipolar to me." dmtallcrazy	Phys Rehab Doc: "Warum nehmen Sie Lamictal?" Ich: "Bipolare Störung." Doc: "Sie wirken nicht bipolar auf mich." dmtallcrazy
2024-02-17 22:13:48.796 - INFO - Translation complete.	

Figure 2: LLM Translation: Original (EN) vs. Translated (DE)

- **Medical Terminology Preservation:** The LLM reliably preserved drug names such as *Paxil*, *Neurontin*, and *Lamictal*. Domain-specific expressions (e.g., "premature ejaculation") were translated accurately as *vorzeitiger Samenerguss*, indicating strong handling of clinical terminology.
- **Register and Style Shift:** Informal social media language was frequently normalized into more formal German. For example, conversational phrases like "I went from awesome vision" were rendered as *Ich ging von einer ausgezeichneten Sicht*, which is grammatically correct but stylistically more formal than typical user-generated content. This suggests a tendency of the model to produce standardized written German.
- **Literal Translation Artifacts:** Some colloquial expressions were translated too literally, resulting in unnatural phrasing (e.g., "Prolia arms" → *Prolia-Armen*). Such cases indicate difficulty in interpreting idiomatic or context-dependent expressions common in social media text.
- **Discourse Marker Adaptation:** Informal markers such as "ya know?" were localized to German equivalents like *oder?*. While semantically appropriate, these substitutions sometimes altered the conversational tone and nuance of the original utterance.
- **Structural Consistency:** Non-linguistic metadata (e.g., [USER] tags) and sentence boundaries were consistently preserved, demonstrating robustness in handling annotated conversational data.

Overall, the translations preserved meaning and medical content reliably but exhibited systematic stylistic normalization and occasional literal translation errors, which may affect downstream modeling on informal social media data.

2.3 Task 1.3: Multilingual and Cross-Lingual Modeling

To ensure methodological stringency and reproducibility, a standardized configuration pipeline was utilized, with parameters programmatically generated via a `generate_config` script to ensure all models were trained under identical baseline conditions. This allowed for an objective comparison between monolingual, multilingual, and translation-based approaches.

Hardware Constraints and Training Environment: Due to the absence of GPU acceleration, all experiments were conducted exclusively on CPU.

This hardware limitation fundamentally shaped the training strategy:

- **Training Duration:** Processing times were significant; monolingual English epochs averaged ~190s, while joint multilingual training exceeded 314s per epoch. Training a single model for 5 epochs typically required 20–30 minutes of active computation.
- **Stability Mechanisms:** To mitigate slow convergence, warmup steps and learning rate scheduling were integrated into the pipeline.
- **Optimization:** **AdamW** served as the primary optimizer across experiments, while **Adafactor** was utilized in specific scenarios to manage memory and stability.
- **Input Specification:** A `max_length` of 256 tokens with truncation and padding to "max_length" was enforced. This specific limit was selected due to observations in the analysis where the maximum token counts reached as high as 1107 in the training set. While the median length remained low across languages, the 256-token threshold was necessary to preserve context in longer German forum posts, which averaged significantly higher lengths (~100 tokens) compared to short English tweets.

Addressing the Accuracy Paradox. Initial training iterations revealed an "Accuracy Paradox" where models achieved > 90% accuracy but yielded an F1-score of 0. Because only 8.1% of the training data was positive, models optimized for the majority class and ignored rare Adverse Drug Event (ADE) cases. To mitigate this, a **weighted loss strategy** was implemented, utilizing the square root of class weights to penalize minority class errors more heavily.

Experimental Iterations: The report discusses the two most significant setups identified during fine-tuning:

- **Experiment A (Downsampled Baseline):** The negative (majority) class was **downsampled by 50%** to create a more balanced training signal, utilized *Dropout* to prevent overfitting, and evaluated thresholds at 0.30 and 0.25.
- **Experiment B (Full-Distribution Optimized):** Utilized the **full training dataset** without downsampling, disabled

Dropout to allow maximum feature extraction, and implemented a more aggressive threshold of **0.15**.

Training Setups and Validation Metrics. Following the task requirements, the performance across the training setups is summarized in Table 3.

Setup	Exp	Lang	Acc	R	F1	Time
Mono	A	en	78.87	0.9141	0.3738	1.1k s
	B	en	94.64	0.6641	0.6308	9.9k s
Mono	A	ru	80.22	0.8565	0.4666	2.8k s
	B	ru	93.03	0.6528	0.6543	23.6k s
Multi	A	EN+RU	83.33	0.8496	0.4513	3.1k s
	B	EN+RU	92.34	0.7097	0.5993	~11k s
Transl.	A	de	5.41	1.0000	0.1026	1.2k s
	B	de	20.95	1.0000	0.1203	1.5k s

Table 3: Validation summary (A: downsampled; B: full-data optimized).

The transition to Experiment B demonstrated that for the full dataset, a threshold of **0.15** provided a superior balance, effectively doubling F1-scores for monolingual models. While the 0.15 threshold improves F1-scores, it results in flagging many normal cases as positive. In the medical domain, this trade-off is often acceptable as the cost of a “False Negative” (missing a side effect) is typically much higher than the cost of a “False Positive” (manual review).

2.4 Task 1.4: Evaluation, Error Analysis, and Cross-Lingual Comparison

As demonstrated in Table 3, Experiment B consistently outperformed the initial baseline. By utilizing the complete training dataset and applying a calibrated decision threshold of 0.15, the system achieved the necessary sensitivity to detect rare Adverse Drug Events (ADE).

This section presents the final evaluation on the held-out test sets, specifically focusing on the model’s ability to generalize to an unseen language (German).

2.4.1 Evaluation Results Summary: The table 4 summarizes the performance across all training configurations. Notably, the **Multilingual (EN+RU)** model was evaluated on German in a zero-shot capacity, while the monolingual models were also tested on German to establish a cross-lingual baseline.

Setup	Lang	Acc	P	R	F1
Mono EN	EN	95.03	0.6232	0.7049	0.6615
Mono RU	RU	92.32	0.6245	0.6176	0.6211
Multi (EN+RU)	EN	94.46	0.5750	0.7541	0.6525
Multi (EN+RU)	RU	92.28	0.6100	0.6728	0.6399
Mono EN	DE	86.75	0.0702	0.1143	0.0870
Mono RU	DE	70.82	0.1287	0.7429	0.2194
Multi (EN+RU)	DE	70.03	0.1256	0.7429	0.2149
MT (EN→DE)	DE	19.56	0.0510	0.7714	0.0957

Table 4: Final metrics on seen vs. unseen (DE) test sets.

2.4.2 Performance Comparison.

- **Monolingual vs. Multilingual:** The **Multilingual (EN+RU)** model is highly competitive, matching native performance while providing superior recall for the Russian subset compared to its monolingual counterpart (67.28% vs. 61.76%).
- **Seen vs. Unseen (Zero-Shot):** A stark contrast is observed in the zero-shot capabilities of monolingual models. The **Mono English** model exhibits “cross-lingual blindness,” failing to recognize German symptoms. However, the **Mono Russian** model performs surprisingly well on German, matching the **Multilingual (EN+RU)** model’s recall (74.29%). This suggests that the Russian dataset provided richer semantic and structural variety for ADE descriptions that mapped effectively to German.
- **Native vs. Machine-Translated:** Native zero-shot transfer (from either the Russian or Multilingual models) is vastly superior to training on a small, synthetic **Machine-Translated** dataset. The translated model suffered a total accuracy collapse (19.56%), becoming a “biased” classifier that over-predicted the positive class.

2.4.3 Cross-Lingual Error Analysis: An analysis of the test data composition explains the performance variations:

- **The English Failure (Tweet-only):** The **Mono English** test set consists exclusively of **Tweets (885)**. This short-form syntax did not prepare the model for the German test set, which features long-form forum sentences and reviews.
- **The Russian/Multilingual Success (Mixed Domain):** The Russian test data includes a diverse mix: **Tweets (1,636)**, **Sentences (934)**, and **Reviews (99)**. This structural variety allowed the Russian-trained models to act as a “**domain bridge**,” successfully transferring knowledge to the similarly structured German forum posts.

The confusion matrices (See Appendix A) visualize these failure modes:

- **Mono English on DE (Zero-Shot Failure):** Exhibits high **False Negatives (31)**, identifying only 11.4% of ADE cases.
- **Mono Russian/Multilingual on DE (Recall Breakthrough):** Both identify **26 out of 35** cases. While **False Positives** increase due to the **0.15 threshold**, the **False Negatives** drop to **9**, significantly improving clinical safety.
- **Machine-Translated on DE (Accuracy Collapse):** Suffers from total predictive collapse (**False Positives: 502**), likely over-fitting to synthetic data artifacts.

2.4.4 Discussion and Practical Implications.

- **Multilingual Training Benefits:** Jointly training on English and Russian creates a language-agnostic understanding that facilitates high-recall zero-shot transfer, especially when source languages provide domain variety (e.g., reviews and sentences).
- **Transfer Limitations:** While clinical entities transfer well, the model remains sensitive to **platform/domain shifts** (e.g., Tweet-to-Forum).
- **Practical Implications:**

- (1) **Recall > Precision:** In medical NLP, high recall is prioritized to ensure rare adverse reactions are not missed.
- (2) **Domain Variety:** Diverse text types are more valuable for robustness than large volumes of a single type.
- (3) **Native Over Synthetic:** Native multilingual transfer is more reliable than small-scale machine translation.

3 Task 2: Classical ML vs LLMs on Structured Data

In this task, different modeling paradigms are compared on a small, fully structured dataset to examine how classical machine learning methods, Transformer-based models, and Large Language Models (LLMs) perform on tabular data. The objective is not only to evaluate predictive performance but also to analyze differences in interpretability, robustness, sample efficiency, and practical applicability.

The experiments are conducted on the **Iris Flower Dataset**, a widely used benchmark for classification of three flower species based on four numerical measurements (sepal length, sepal width, petal length, and petal width). The dataset was randomly split into **80% training**, **10% validation**, and **10% test** sets to ensure fair evaluation across all approaches.

Task 3.1 builds a Decision Tree baseline, Task 3.2 applies Transformers to textified data, Task 3.3 evaluates LLM-based prediction, and Task reftask3.4compares all approaches in terms of performance, interpretability, and robustness.

3.1 Classical Baseline on Structured Data

A Decision Tree was selected as the classical baseline primarily for its interpretability and transparency. Given that Task 2.3 requires augmenting an LLM with structured information such as decision rules and feature importance, a Decision Tree provides an accessible 'white-box' logic that other models lack. Furthermore, the tree's ability to perform automatic feature selection provided immediate insight into the characteristics of the Iris dataset.

3.1.1 **Model Behavior and Hyperparameter Constraints.** Distinct behaviors were observed between the unconstrained and constrained Decision Tree models:

- **Overfitting in the Unconstrained Model:** Initially, without a depth limit, a training accuracy of 100% was achieved, while the test accuracy was limited to 87%. This discrepancy indicates overfitting, where specific training instances were memorized rather than generalizable rules being learned.
- **Improved Generalization with Pruning:** By enforcing a constraint of *max_depth* = 3, the model was forced to focus on the most significant patterns. Although training accuracy was slightly reduced to 98.3%, the test accuracy improved to 93%, demonstrating superior generalization to unseen data.

3.1.2 **Tree Logic and Feature Importance.** The visual structure of the model is shown in Figure 3. This diagram illustrates how the model makes decisions at each step.

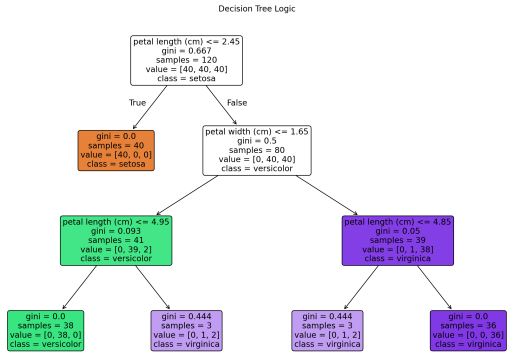


Figure 3: Decision Tree Visualization (Max Depth = 3)

As seen in Table 5, petal length and petal width were identified as the only features used for classification. Sepal length and sepal width received importance scores of 0.00, which shows the decision logic relied entirely on petal dimensions. It was determined that sepal measurements were redundant, as the model performed automatic feature selection.

As seen in Table 5, petal length and petal width were identified as the only features used for classification. Sepal length and sepal width received importance scores of 0.00, which shows the decision logic relied entirely on petal dimensions. It was determined that sepal measurements were redundant, as the model performed automatic feature selection.

Feature	Importance
Petal length (cm)	0.557
Petal width (cm)	0.442
Sepal length (cm)	0.000
Sepal width (cm)	0.000

Table 5: Calculated Feature Importance for the Decision Tree model.

3.1.3 **Strengths and Limitations.** The following observations were made regarding the model's performance:

- Strengths:**
- **Interpretability:** The model is characterized as a "white-box," ensuring that the classification logic is transparent.
 - **Efficiency:** It requires minimal data preprocessing (like scaling) and is computationally very fast to train on small datasets.
 - **Feature Selection:** It naturally identifies and prioritizes the most discriminative features (e.g., petal dimensions).

Limitations:

- **Sensitivity to Depth and Overfitting:** Without constraints like *max_depth*, the model tends to memorize the training data, leading to poor performance on unseen test data.
- **Instability:** Small changes in the training data can result in a completely different tree structure, making it less robust than some ensemble methods.

3.2 Textification and Transformer-Based Modeling

In this section, the structured data was converted into different text representations to allow the Transformer model to process numerical measurements. Two specific formats were tested and compared:

3.2.1 Representation A: Full sentences to describe the measurements.

- **Example:** The flower has a sepal length of 6.8cm, a sepal width of 3.0cm, a petal length of 5.5cm, and a petal width of 2.1cm.
- **Justification:** Transformers are pre-trained on massive amounts of natural text, such as books and Wikipedia. Using full sentences might make it easier for the model to understand the context of the numbers.

3.2.2 Representation B: A short and structured format with labels.

- **Example:** sl:6.8 sw:3.0 pl:5.5 pw:2.1
- **Justification:** This format is more token-efficient. In real-world NLP, space is often limited, and this compact style allows the model to reach a conclusion using less information.

3.2.3 Transformer Training. The **DistilBERT** model was fine-tuned in several experiments using different learning rates, with the Macro F1-score used to select the best model. The results suggest that the model could be sensitive to training conditions. **Although seeds were set**, different training samples or runs might have led to varying results. This could indicate possible instability on small datasets, where performance sometimes remained similar across representations and sometimes appeared to favor **Representation A**.

The performance comparison for both runs is summarized in Table 6. As seen in Table 6, **Representation B** reached a high training Macro F1-score of 97.5%. However, test performance varied between runs. In Run 2, **Representation A** reached a superior test Macro F1-score of 93.3%, while Representation B maintained 97.5% in training but dropped to 86.6% on the test data. This variability could suggest that while the compact format might help the model learn the training set effectively, the natural language format could potentially generalize better to unseen data in certain cases.

Exp.	Rep.	Train F1	Val F1	Test F1
Exp. 1	A (Natural)	0.975	0.933	0.866
	B (Compact)	0.975	0.933	0.866
Exp. 2	A (Natural)	0.950	0.933	0.933
	B (Compact)	0.975	0.933	0.866

Table 6: Comparative Metrics for Transformer Training Runs

3.3 LLM-Based Prediction and Hybrid Modeling

Large Language Models (LLMs) were used to classify Iris flower measurements. Representation B was selected for all experiments due to its compact structure, which reduces token usage and improves inference efficiency.

3.3.1 Model Selection. Experiments were conducted using locally hosted models and the Academic Cloud, including **Mistral Devstral (24B)** and **Mistral Large (675B)**. These models were chosen for their strong instruction-following behavior and ability to produce structured outputs. Other models, such as standard GPT or smaller open-source models, often produced extra text or explanations even when instructed to return only JSON.

3.3.2 Prompt Design. Two prompt configurations were evaluated:

- **Baseline Prompt:** Relies on the model’s internal knowledge with a strict JSON output schema to ensure machine-readable predictions.
- **Augmented Prompt:** Incorporates structured knowledge from a classical model, including decision rules and feature importance, instructing the LLM to prioritize these constraints during reasoning.
- **Few-Shot Guidance:** Three labeled examples per class were included to demonstrate the input format and expected behavior.
- **Output Guardrails:** Instructions such as “no explanations” and “JSON only” prevented conversational responses that would break automated evaluation.

3.3.3 Inference Workflow and Validation. The inference pipeline consisted of the following steps:

- (1) **Input Encoding:** Numerical features were converted into Representation B.
- (2) **Context Construction:** System instructions, optional rules, and few-shot examples were combined into a single prompt.
- (3) **LLM Inference:** The model generated a structured prediction.
- (4) **Output Validation:** Responses were validated to ensure correct JSON format and presence of required fields (label and confidence). Invalid outputs were discarded to maintain evaluation integrity.
- (5) **Metric Computation:** Valid predictions were used to compute accuracy and F1 scores.

Model	Split	Acc.	F1	time(s) / Sample
LLM Only	Train	0.9369	0.9369	15.88
	Val	0.9333	0.9327	11.70
	Test	0.9333	0.9327	12.59
Hybrid	Train	0.9279	0.9284	13.38
	Val	1.0000	1.0000	13.00
	Test	0.8667	0.8667	13.31

Table 7: LLM Baseline vs. Hybrid Performance

3.3.4 Comparison and Conclusion. The baseline LLM (few-shot only) achieved the best overall generalization, with a test Macro F1 of 0.9327 compared to 0.8667 for the hybrid model. While

the hybrid approach reached perfect validation performance, its test accuracy dropped, indicating reduced robustness to unseen samples. Per-class results show that the hybrid model performed perfectly on *setosa* but degraded on the more ambiguous boundary between *versicolor* and *virginica*.

Augmenting the LLM with classical decision rules and feature importance did not improve performance for this task and slightly increased inference time. Few-shot prompting alone provided a better balance of accuracy and generalization. These results suggest that, for simple tabular classification problems, additional symbolic constraints may over-constrain the model and interfere with its internal pattern-based reasoning.

3.4 Comparative Analysis and Reflection

Approach	Train	Val	Test
DT	0.983	1.00	0.933
TF-A	0.975	0.933	0.867
TF-B	0.975	0.933	0.867
LLM	0.937	0.933	0.933
LLM+R	0.928	1.00	0.867

Table 8: Comparison on Iris (Macro F1).

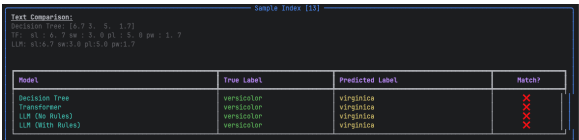
3.4.1 . Performance Differences. The strongest test performance was achieved by the Decision Tree and the LLM (no rules), both reaching a Macro F1 of 0.933. The Transformer variants underperformed on the test set (0.867), suggesting limited generalization from textified inputs under the given training setup. Adding structured guidance (rules and feature importance) to the LLM did not improve predictive performance (Macro F1 = 0.867) and additionally increased inference time. This indicates that the added constraints neither improved accuracy nor efficiency, and may have introduced unnecessary complexity, particularly in ambiguous regions of the feature space.

3.4.2 Sample Efficiency. The Decision Tree is highly sample-efficient for this task: it learns strong decision boundaries from few numeric samples. The LLM approach requires no gradient-based training and can perform competitively with only a small number of in-context examples (3-shot). In contrast, the Transformer requires supervised training and tuning, making it the least sample-efficient and most costly to iterate on.

3.4.3 Interpretability. The Decision Tree is the most interpretable. Feature importances show that decisions rely almost entirely on petal length and petal width, while sepal features contribute negligibly. The hybrid LLM is partially interpretable due to explicit rules being present in the prompt, but the final prediction can still deviate from these constraints. The Transformer and LLM-only approaches provide weaker interpretability, as their decision logic is implicit.

3.4.4 Robustness and Consistency. Across the inspected test samples, all models failed on the same difficult case (Sample 4: true *versicolor* predicted as *virginica*), suggesting an intrinsically ambiguous region. The hybrid LLM introduced an additional error (Sample

5: true *virginica* predicted as *versicolor*) that the Decision Tree and LLM-only handled correctly, indicating increased brittleness under prompt constraints.



Model	True Label	Predicted Label	Match?
Decision Tree	versicolor	versicolor	✓
Transformer	versicolor	virginica	✗
LLM (No Rules)	versicolor	versicolor	✓
LLM (With Rules)	versicolor	virginica	✗

Figure 4: Sample-level comparison across models.



Model	True Label	Predicted Label	Match?
Decision Tree	virginica	virginica	✓
Transformer	virginica	versicolor	✗
LLM (No Rules)	virginica	virginica	✓
LLM (With Rules)	virginica	versicolor	✗

Figure 5: Sample-level comparison across models.

3.4.5 Practical Recommendations.

- **Decision Tree:** Demonstrates strong accuracy, fast training and inference, and high interpretability through explicit rules and feature importance, making it well-suited for small tabular classification tasks.
- **LLM (few-shot, no rules):** Provides competitive performance without requiring model training, making it a practical option when rapid deployment is needed or when inputs are available in natural-language or compact textual form.
- **Rule-augmented LLM prompting:** In this setup, the addition of structured rules and feature importance did not improve generalization and slightly reduced robustness on unseen data, indicating that such augmentation may require careful validation.
- **Transformer on textified data:** Shows potential for scalability and handling inherently textual inputs, but was less competitive for this small structured dataset compared to simpler models.

4 Conclusion

Multilingual ADE detection in medical social media was evaluated under severe class imbalance and domain shift. Multilingual training on English and Russian enabled effective zero-shot transfer to German, while English-only models showed limited cross-lingual generalization. Training on a small machine-translated dataset proved unreliable compared to native multilingual transfer. These results indicate that domain diversity and recall-oriented optimization are crucial for medical NLP tasks.

On the structured Iris dataset, the Decision Tree achieved strong performance with high interpretability and sample efficiency. Transformer models on textified data showed unstable generalization, whereas few-shot LLM prompting achieved competitive results without training. Overall, model effectiveness was found to depend strongly on data characteristics, with simpler or appropriately prompted methods often outperforming more complex approaches in low-data settings.

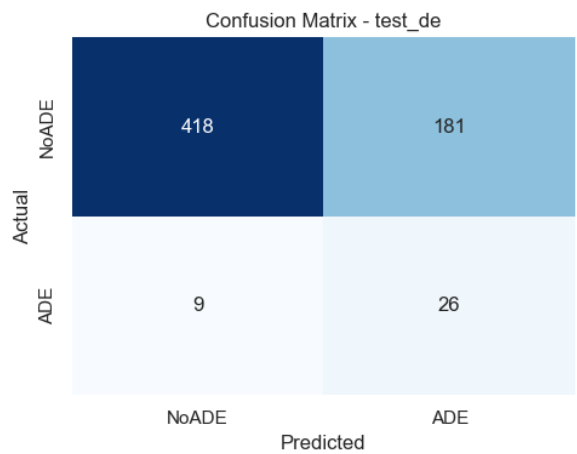


Figure 8: Multilingual (EN+RU) model evaluated on German test set.

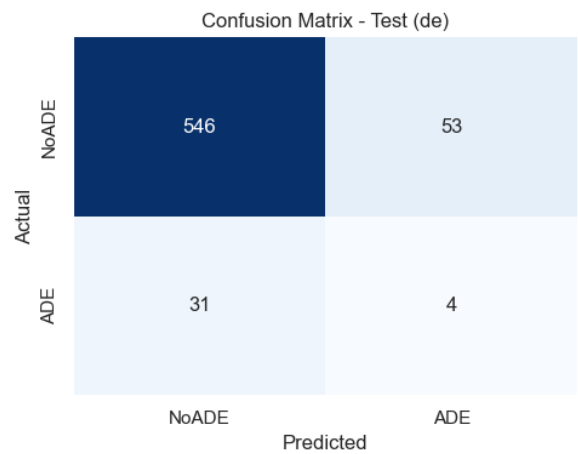


Figure 6: Mono English model evaluated on German test set.

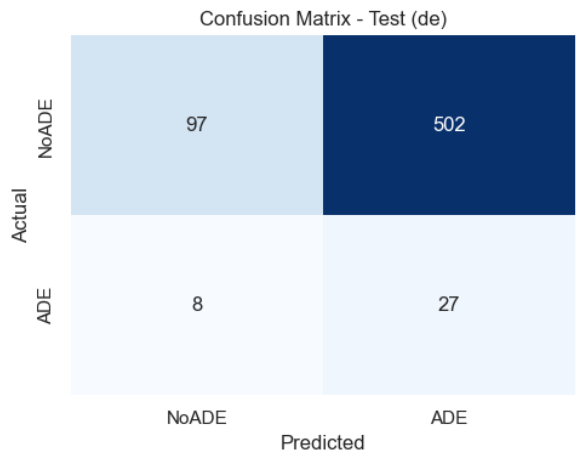


Figure 9: Machine-Translated English-to-German model.

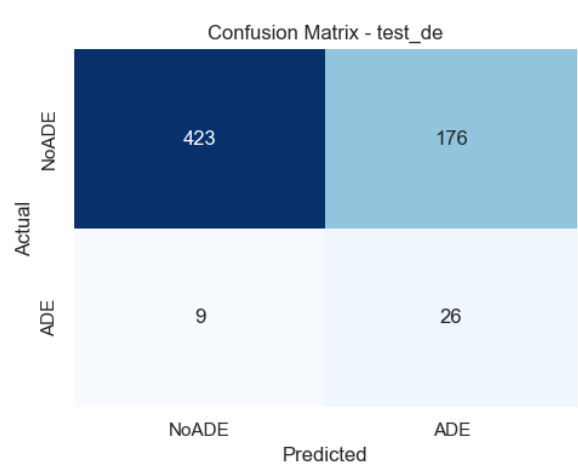


Figure 7: Mono Russian model evaluated on German test set.

A Confusion Matrices (German Test Set)

The following figures illustrate the classification performance on the unseen German test set for each experimental setup in Task 1.